

The steeper 1/f slope of MEG power spectrum is associated with stronger gamma synchrony, measured via inter-trial phase coherence (ITPC) in the 40-Hz auditory steady-state response

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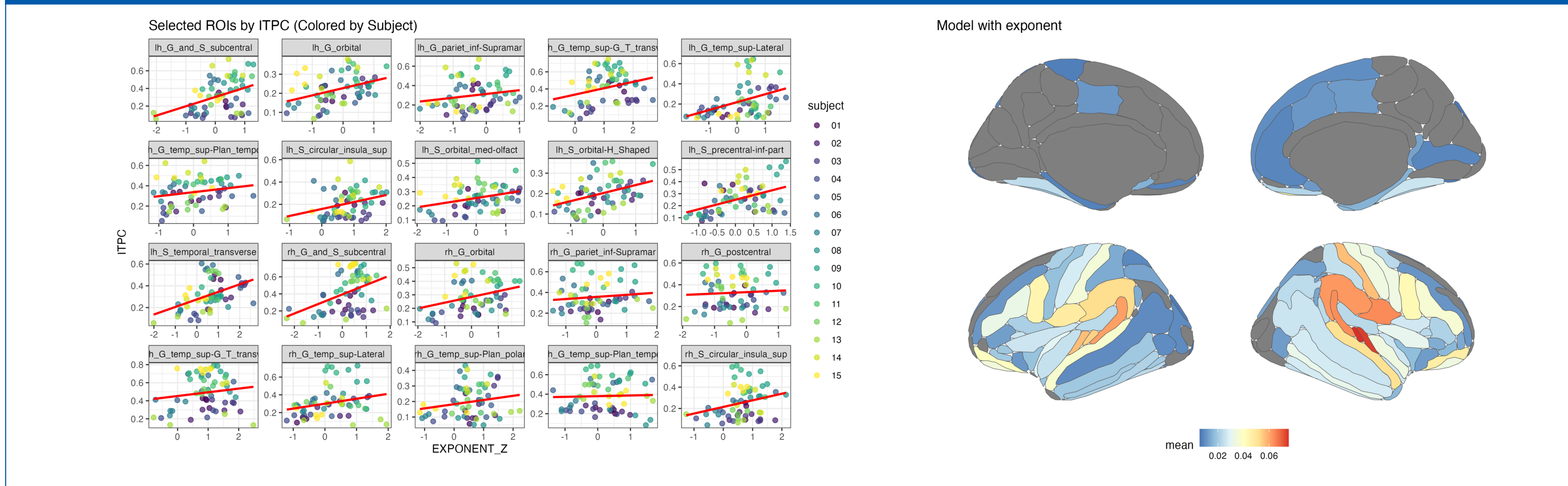
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Hypothesis. We explored whether a steeper slope of the 1/f PSD component (i.e., a larger spectral exponent) in MEG signals is associated with higher auditory steady-state response (ASSR) synchrony, as measured by inter-trial phase coherence (ITPC) during 40 Hz auditory stimulation.

Introduction

- Power spectral density (PSD) of electrophysiological activity may be parametrized by oscillatory components and aperiodic component, where the power of the signal decay with a 1/f power law
- The timescale and magnitude of this aperiodic activity can be described by the slope and offset of this 1/f decay [2]

Relationship between the slope (parametrized by exponent and the estimated ITPC). Figure 2 on the left: Relationship between Spectral Exponent (z-scored) and ITPC across selected ROIs. Red line indicates linear fit. Data points colored by subject.



Methods

- The PSD of the spectrum were computed for 1s second pre-stimulus period and stimulation period for each trial (and averaged later) and for each region. The slope and exponent were computed using specparam toolbox [2].
- ITPC was computed, the value averaged in the intervals [-500, -200] ms for prestimuli and [400, 700] ms after sound onset for each of the regions using the wavelet approach described in [4].
- The influence of the stimulus and the exponent on the ITPC were tested by modelling the different relationships with an integrative Bayesian Multilevel Model [1]
- This methods allows for better integration of the spatial information and control of multiplicity avoding information loss of univariate approach followed by the correction for multiple testing.
- Effect estimation and region-level inferences are based on posterior distributions of group differences.

Following models were tested:

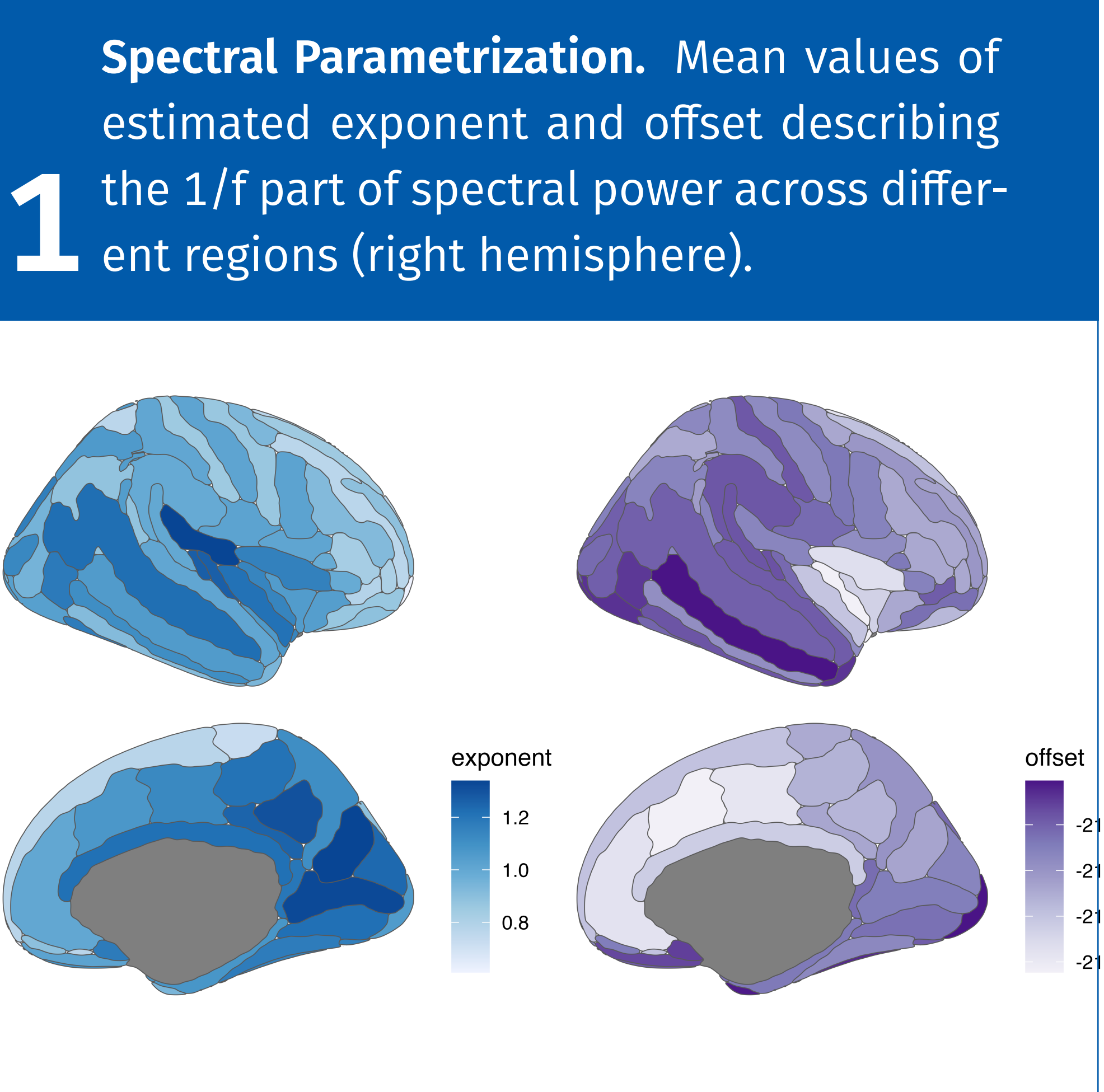
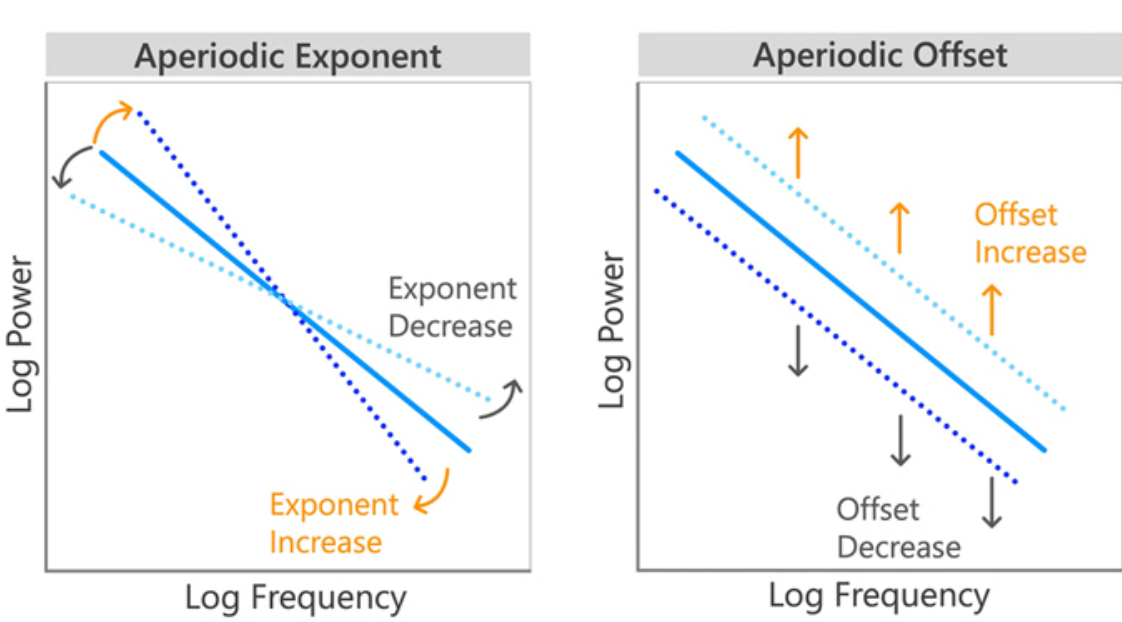
- ITPC ~ S (Stimulus)
- ITPC ~ Exponent + Stimulus
- Exponent ~ Stimulus
- Offset ~ Stimulus

- For investigation of the the relationship we fit the model

$$ITPC_{stim} \sim 1 + exponent_z + (1 + exponent_z | subject) + (1 | ROI)$$

with z-scored Exponent. The model was fitted using the brms package and [1].

- Simplified formula describing 1/f component of the PSD of the signal: $L(f) = b - \log(k + f^\chi)$; The exponent χ is related to the negative slope of the power spectrum in log-log space, that is, the inverse relationship between power and frequency; The offset b describe the uniform shift in power across frequencies
- These parameters are most likely governed by waxing and waning of oscillatory components, and can be associated to biological mechanisms (e.g. excitation:inhibition balance as summed by [7], cognitive processes, and group differences).
- Graphical relationship of spectral power and offset and exponent (from [3])



- Synchronization of brain activity in the gamma frequency is a hallmark of brain function, and an indicator of malfunctioning[4].
- Auditory stimulation in the gamma range is a reliable way to experimentally investigate this phenomenon, looking at auditory steady-state responses (ASSR), the effect of entrainment of brain activity to the frequency and phase of external auditory stimuli.
- ASSR are usually investigated by looking at the inter-trial phase coherence (ITPC) [4].

Data

- We re-analysed MEG data from n=15 healthy participants, stimulated with a sound wave modulated at 40 Hz, presented in 180 trials for one second, with the onset of two consecutive trials separated by at least 2 s [4].
- The data were preprocessed, projected into Destrieux brain atlas

Conclusion

Main Finding:

- Steeper spectral slopes associated with stronger gamma phase-locking, particularly in auditory and temporal regions showing robust ASSR responses

Two Possible Mechanisms:

- Biological:** Both measures may reflect E/I balance—stronger inhibitory tone (steeper slope) provides temporal scaffolding for precise synchronization [7]
- Methodological:** Spectral slope influences phase estimation reliability and SNR [6, 5]
- Regional specificity and exponent stability suggest biological contribution, though methodological confounding cannot be fully ruled out

Implications:

- References**
- Exploratory findings ($n = 15$) warrant replication in larger samples
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